

Learning-Induced Metal-Insulator Transition

Exploratory DIII monitored-Majorana study with an XY-line validation gate

We reproduce the known XY-line learning transition, then test a generic symmetry-class-DIII measurement cut with Born-sampled Gaussian Majorana trajectories. The report explains the mapping, algorithms, parameters, finite-size analysis, anisotropy calibration, uncertainty, and strict exploratory claim gates.

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Executive result and scope

Exploratory status

The frozen status is `xy_reproduced_diii_inconclusive`. The directly fitted universal candidate is `c_eff` α =None; α =None is reported only with its window-stability gate.

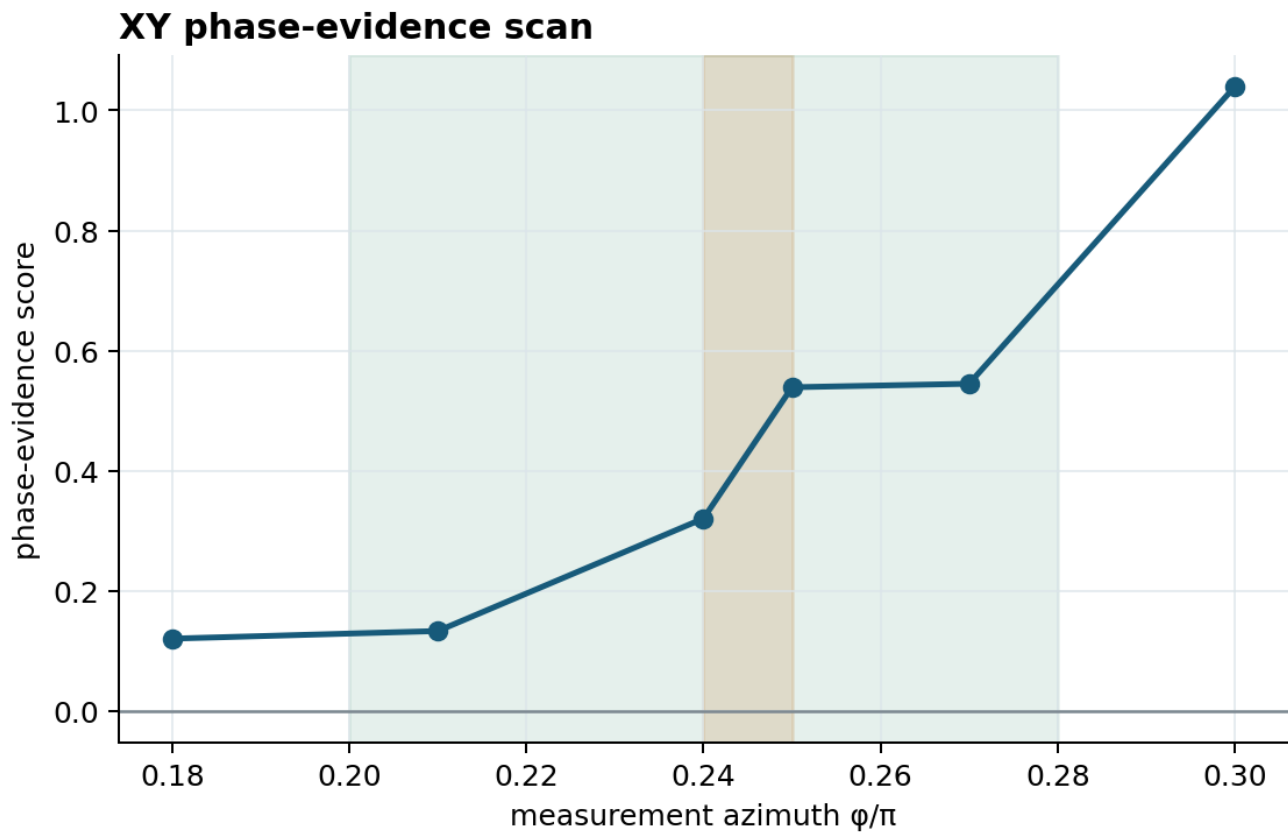


Figure 1. XY validation scan and declared reference window.
Interpretation limit: Finite widths and declared fit windows limit interpretation.

Basic concepts: monitoring and phases

Repeated quantum measurements can either preserve extended Majorana correlations or localize them. The metal-insulator distinction is diagnosed from size trends, entanglement-arc model weights, correlations, and record free energy rather than from one smooth curve.

$$\sigma(\theta, \varphi) = \sin\theta \cos\varphi X + \sin\theta \sin\varphi Y + \cos\theta Z$$

The two angles define the physical single-qubit measurement axis.

Surface-code mapping and effective couplings

The surface-code Born tensor network maps to alternating Gaussian measurements and rotations. The real coupling controls information gain; the complex phase controls norm-preserving Majorana rotation.

$$J(\theta) = \text{atanh}(\cos\theta)$$

This converts the polar measurement angle into a real monitoring strength.

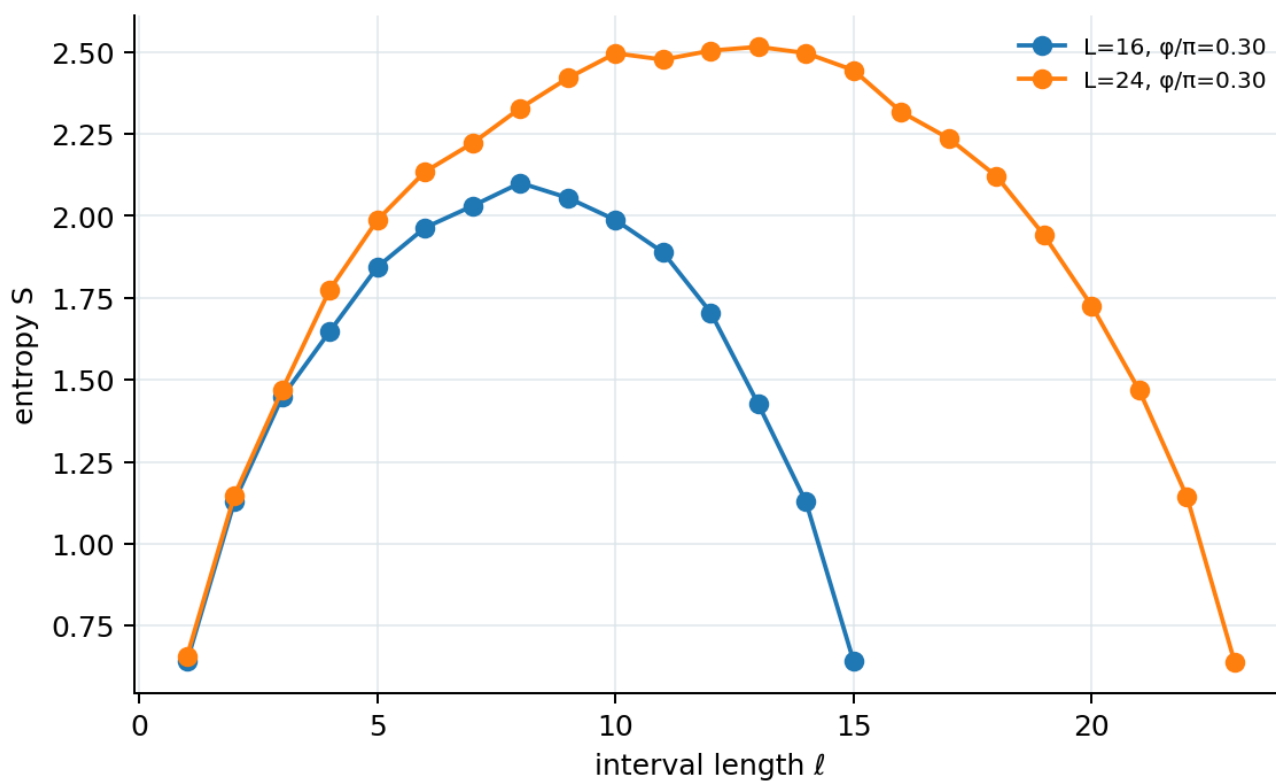


Figure 2. Representative entanglement arcs across angles and widths.
Interpretation limit: Finite widths and declared fit windows limit interpretation.

Why generic DIII differs from the XY class-D line

The XY line has a special class-D block decomposition. Moving to $\theta=0.45\pi$ with nonzero azimuth removes that decomposition and realizes the generic DIII cut targeted by the open challenge.

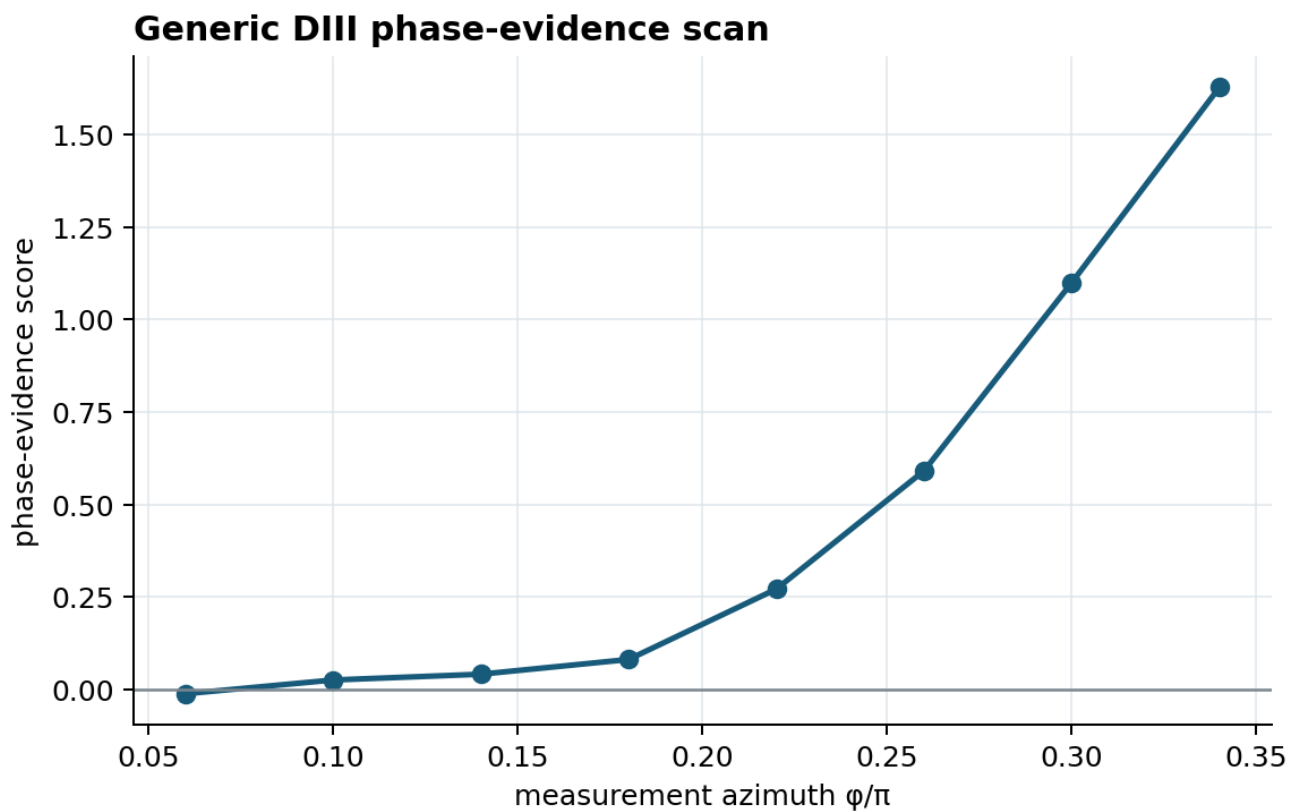


Figure 3. Generic DIII coarse evidence and selected refinement bracket.
Interpretation limit: Finite widths and declared fit windows limit interpretation.

Gaussian covariance implementation

Rust evolves a real antisymmetric $2L \times 2L$ covariance matrix. Rational measurement updates and orthogonal rotations preserve purity analytically; after each period, an orthogonal polar projection removes accumulated roundoff in $\Gamma^2 = -I$. Entropy follows from duplicated singular values of restricted covariance matrices.

$$\Gamma' = R \Gamma R$$

Every unitary Majorana gate acts by congruence on the covariance matrix.

Conditional Born sampling and the IID negative control

Xoshiro256++ draws conditional Born outcomes. The Rao-Blackwellized binary entropy is accumulated before each draw. Unbiased IID signs are a deliberately nonphysical diagnostic and never enter physical summaries.

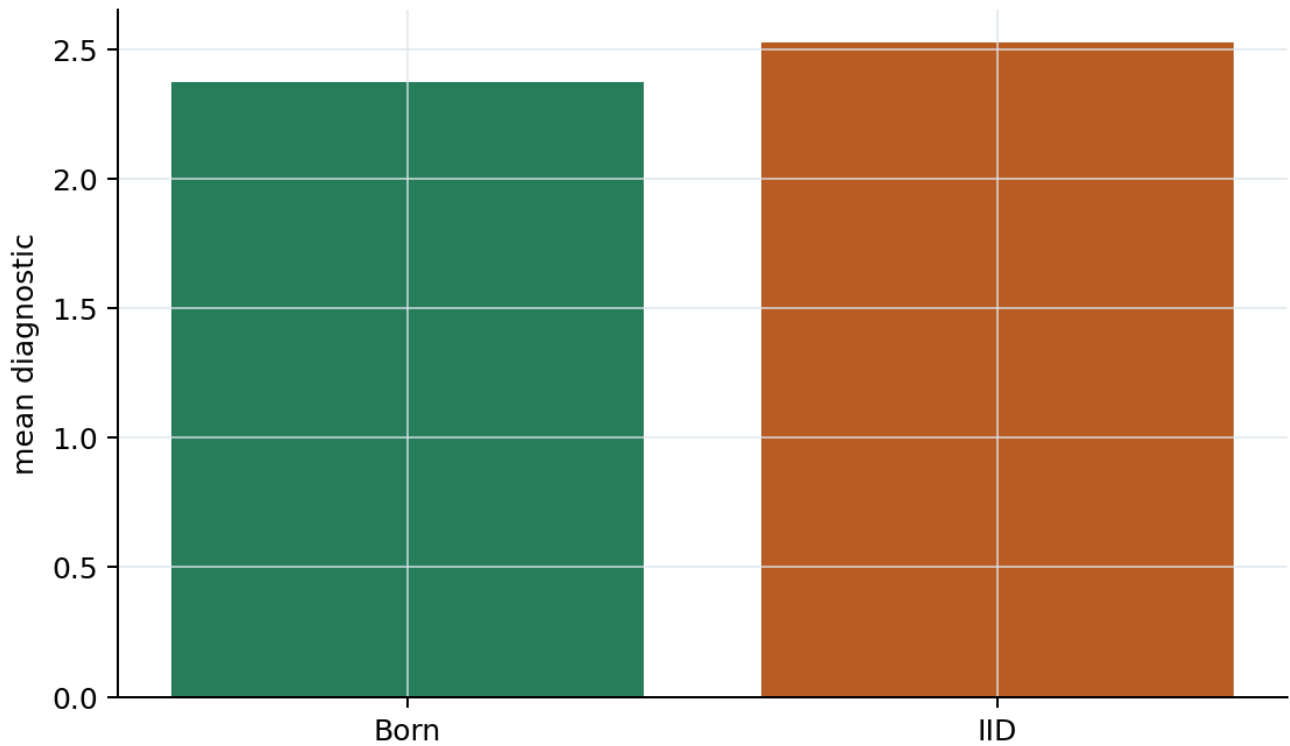


Figure 4. Born sampling compared with the nonphysical IID-sign control.
Interpretation limit: Finite widths and declared fit windows limit interpretation.

Parameters, runtime budget, and estimator meanings

Table 1. Parameter settings

Parameter	Setting	Meaning
Widths L	8, 12, 16, 24	finite-size scaling
Independent streams	4	stream-level uncertainty
Ordinary stop	3300	stop launching new work
Hard stop	5100	atomic finalization only

Times are measured from the production simulation and analysis pipeline.

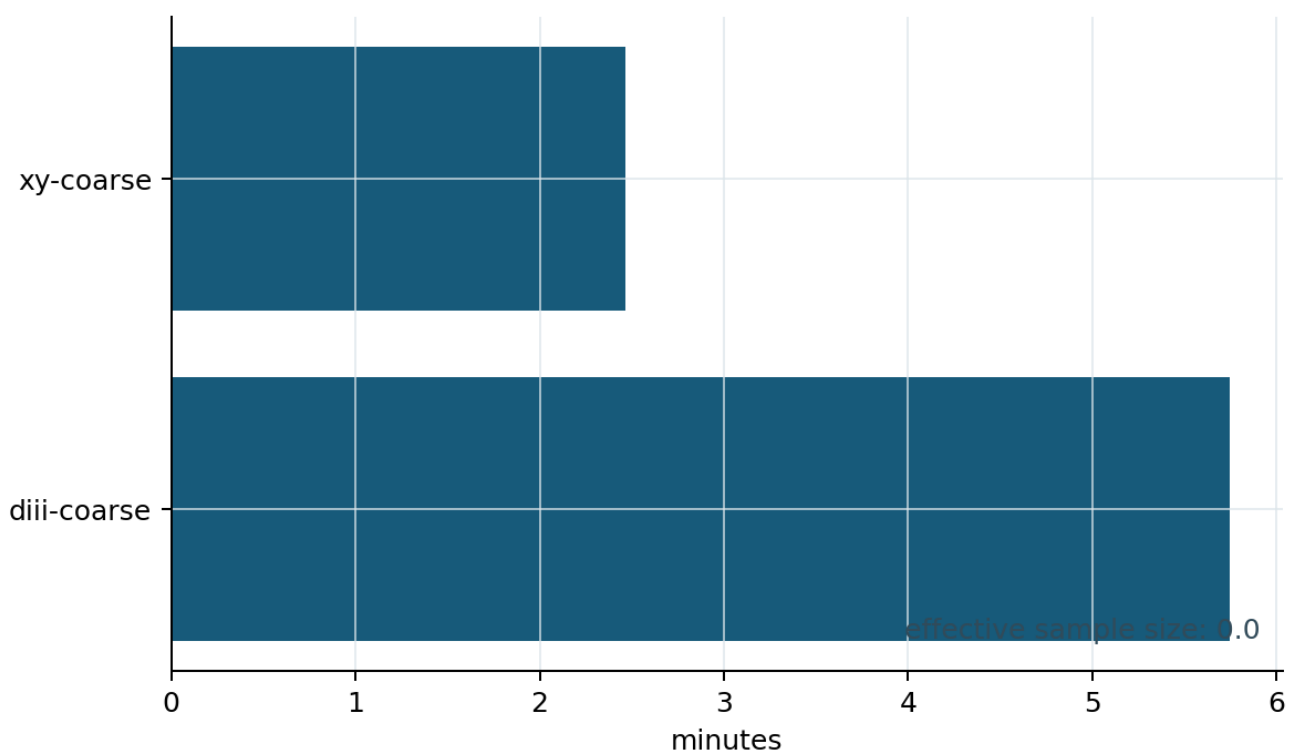


Figure 5. Runtime allocation and effective sample size.

Interpretation limit: Finite widths and declared fit windows limit interpretation.

Independent mathematical and physical oracles

Dense Hilbert-space enumeration at $L=2$ checks joint probabilities, covariances, and entropy independently. Analytic angle limits, the frozen weak-self-dual kernel, exact-Y swaps, and class-D decomposition residuals guard sign and factor conventions.

Table 2. Scientific oracles

Oracle	maximum error
dense Born probability	1.006e-16
dense covariance	1.776e-15
weak self-dual limit	not recorded

Production is gated on these checks.

Reproduction of the known XY transition

The XY scan is a validation gate: its finite-size bracket must overlap the declared reference window before any generic-DIII claim is considered.

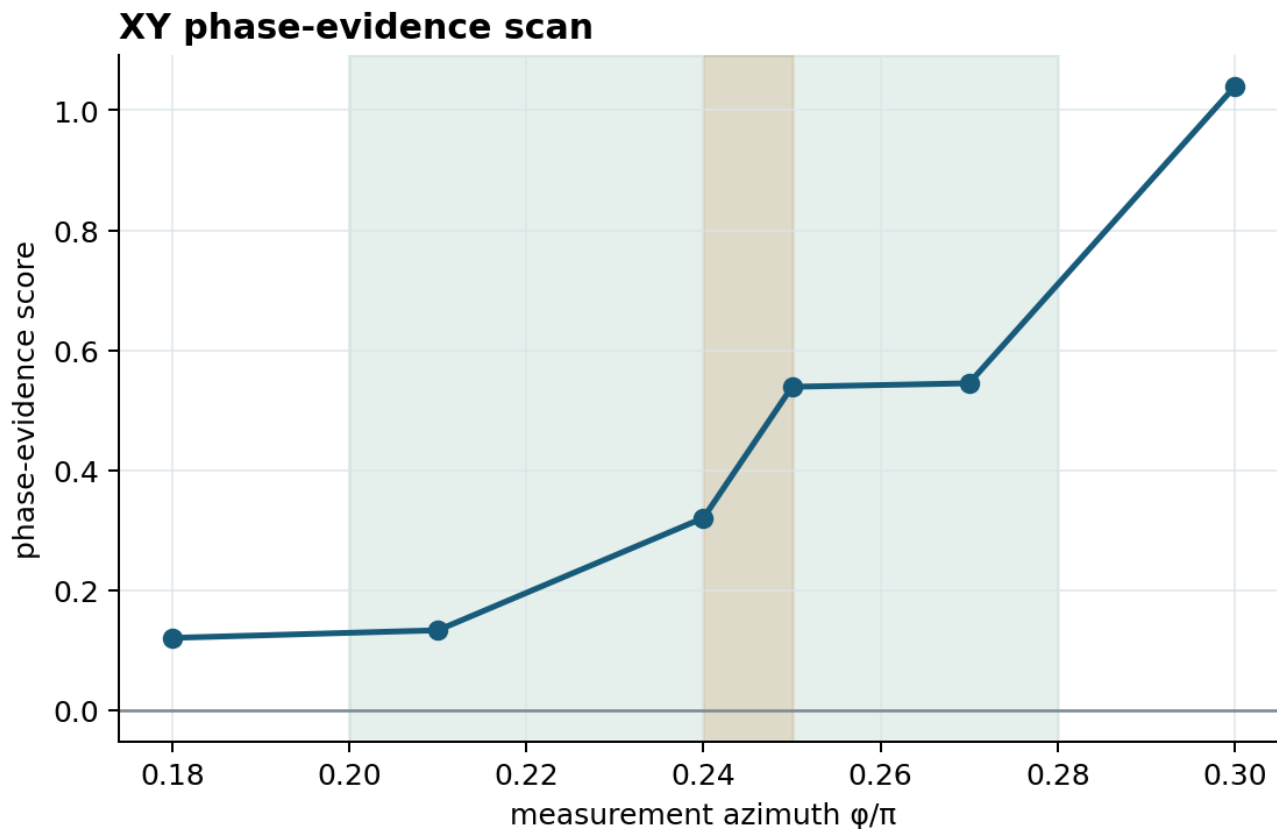


Figure 6. XY validation scan and declared reference window.

Interpretation limit: Finite widths and declared fit windows limit interpretation.

Exploratory generic-DIII phase evidence

The generic scan compares area, logarithmic, squared-logarithmic, and Page-augmented arc models across widths. A transition bracket requires persistent opposite phase evidence on adjacent angles.

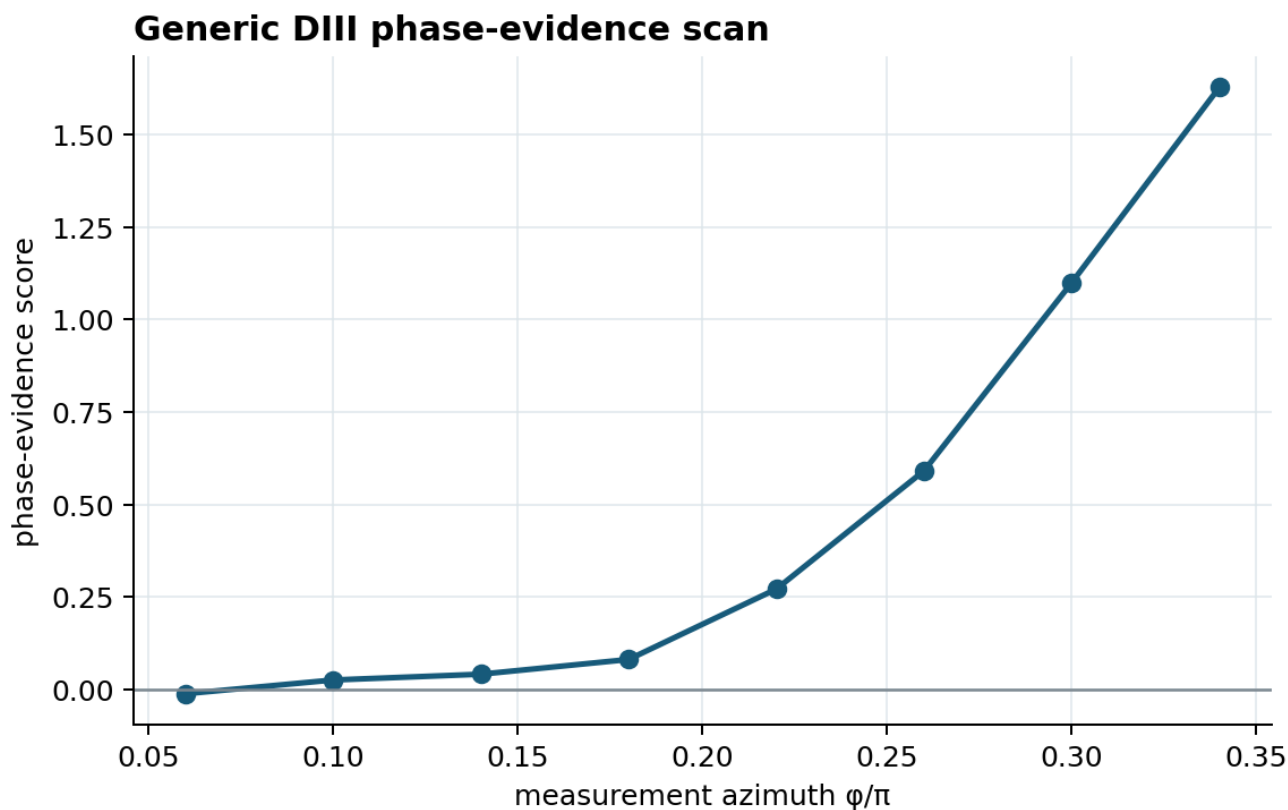


Figure 7. Generic DIII coarse evidence and selected refinement bracket.
Interpretation limit: Finite widths and declared fit windows limit interpretation.

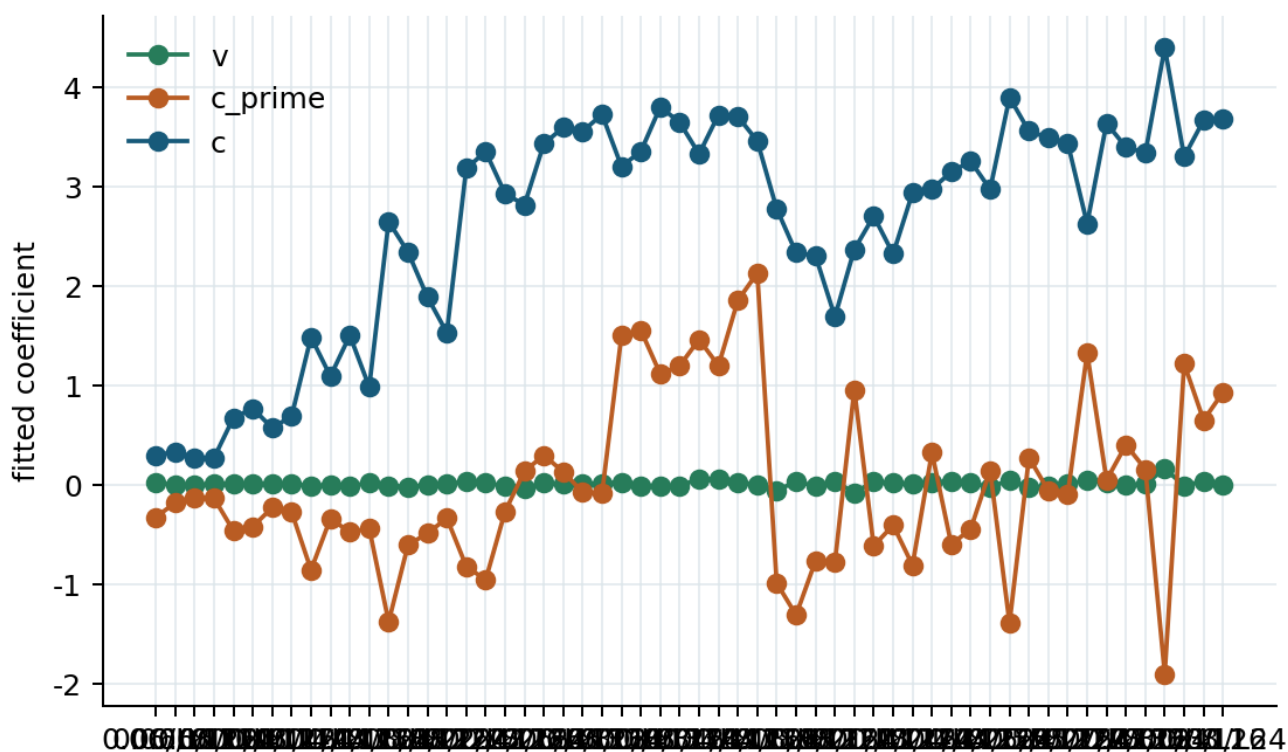


Figure 8. Arc-model coefficients used to distinguish area, logarithmic, and squared-logarithmic behavior.
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Casimir amplitude and finite-size correction

$$\gamma(L) = f_{\infty} L - \pi(c_{\text{eff}} \alpha) / (6L) + a/L^3$$

The coefficient of $1/L$ is the directly observed product $c_{\text{eff}} \alpha$; the L^{-3} term absorbs leading finite-size drift.

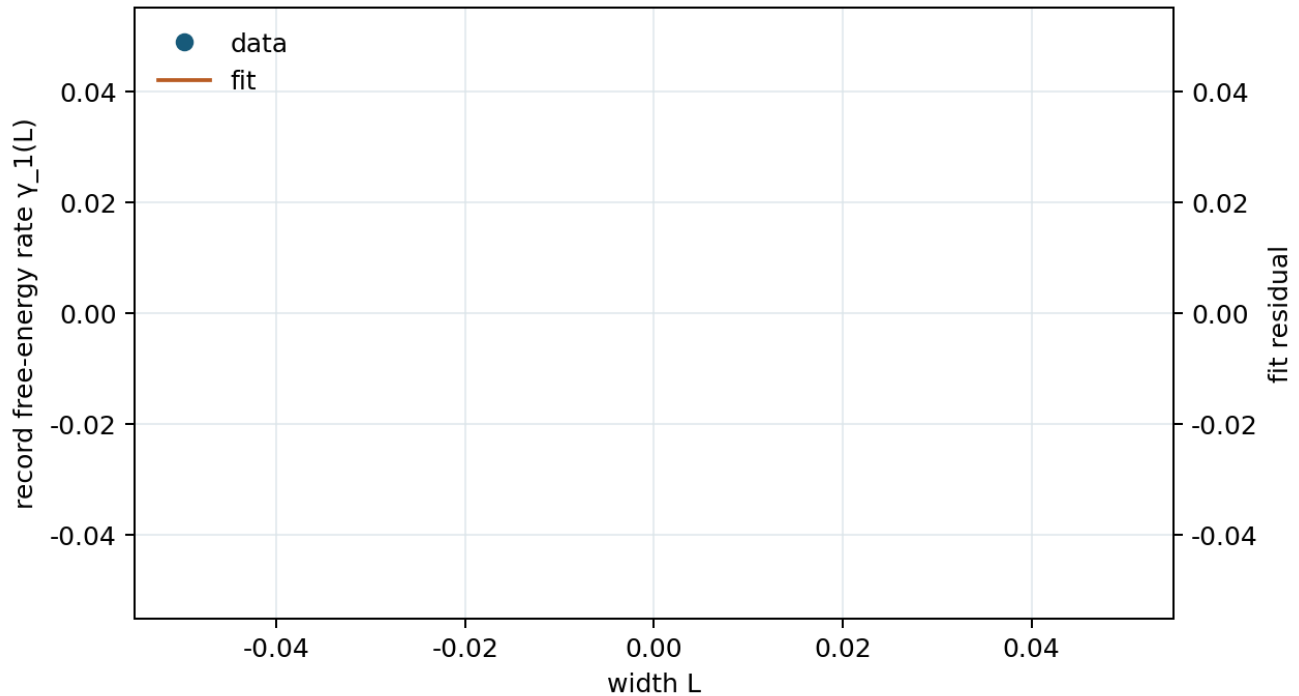


Figure 9. Casimir finite-size fit with residuals.

Interpretation limit: Finite widths and declared fit windows limit interpretation.

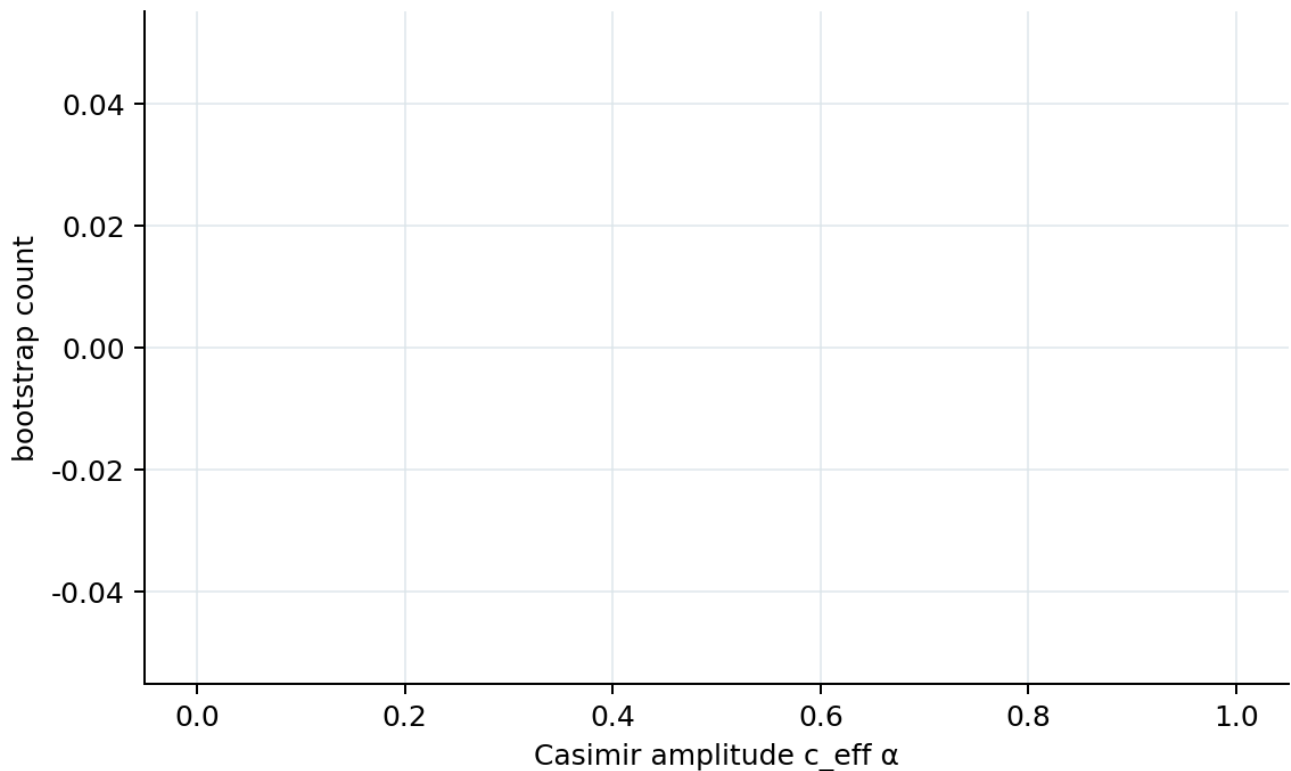


Figure 10. Hierarchical bootstrap distribution of $c_{\text{eff}} \alpha$.

Interpretation limit: Finite widths and declared fit windows limit interpretation.

Spatial-temporal anisotropy calibration

Spatial parity correlations determine Δ , while the leading temporal Lyapunov gap determines α . Window changes and block deletions must agree before dividing the Casimir amplitude by α .

$$\alpha = gL / (2\pi\Delta)$$

This calibration makes the spacetime conversion explicit instead of assuming isotropy.

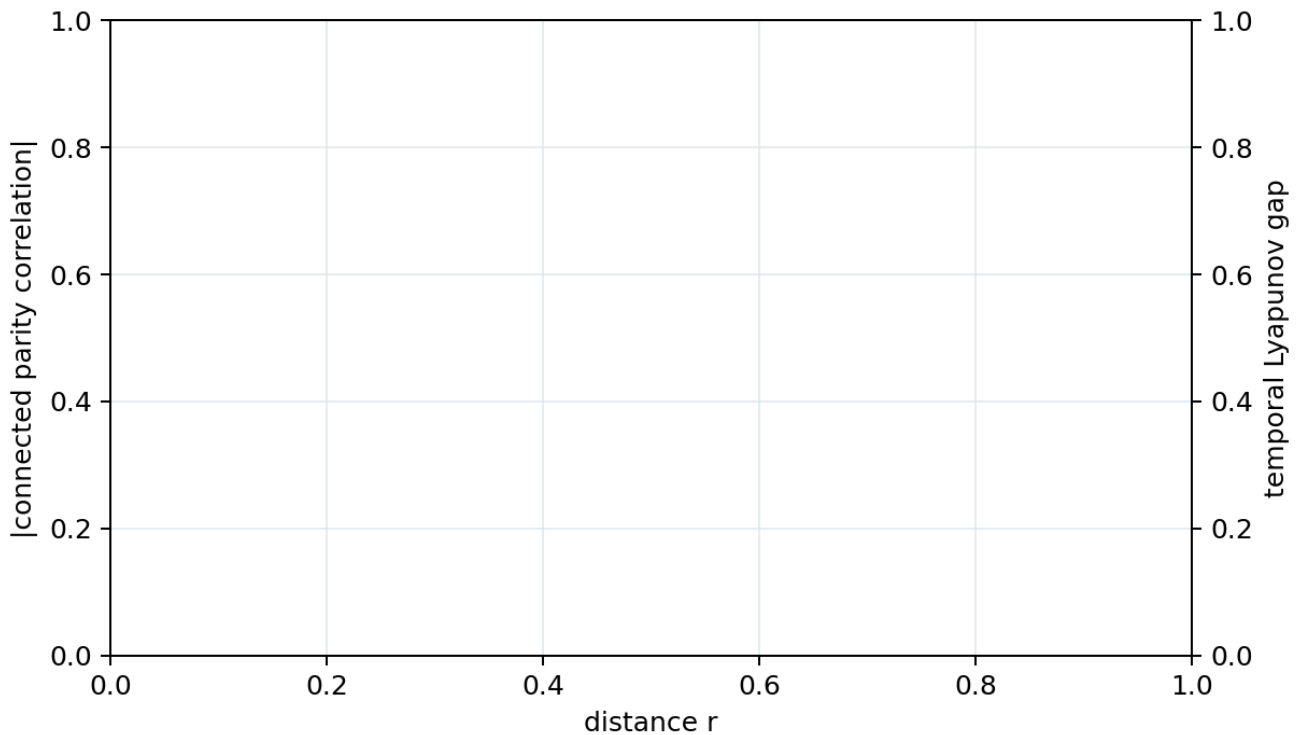


Figure 11. Spatial parity decay and temporal Lyapunov gaps.
Interpretation limit: Finite widths and declared fit windows limit interpretation.

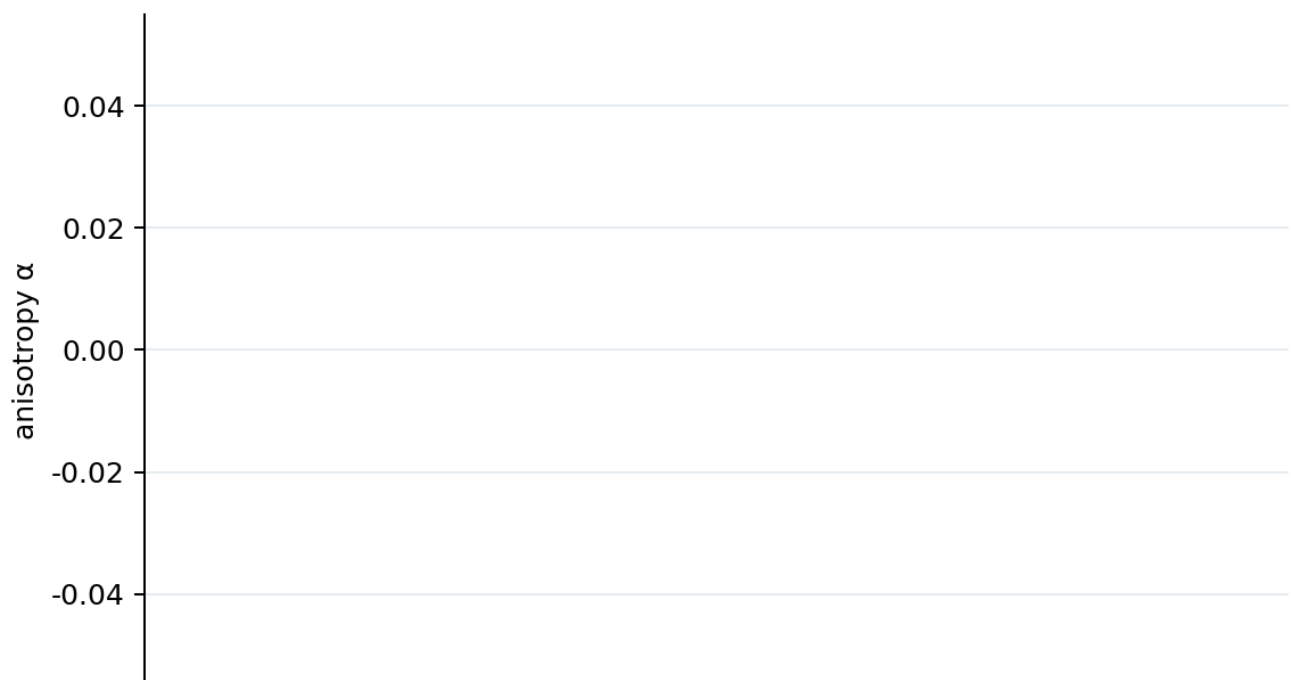


Figure 12. Anisotropy sensitivity to analysis windows.

Interpretation limit: Finite widths and declared fit windows limit interpretation.

Uncertainty, sensitivity, and claim boundary

Uncertainty is hierarchical: streams are resampled first and complete blocks second. Sensitivity checks change minimum width, corrections, phase brackets, correlation windows, and Lyapunov block deletions.

Claim boundary

This is an exploratory finite-size result, not a final universal constant. If a is unstable, the report publishes c_{eff} a only and suppresses standalone c_{eff} .

Reproducibility and code/data inventory

All physics sampling is implemented in Rust; Python only validates frozen artifacts, fits models, bootstraps, plots, and renders reports. Stream JSON, block CSV, refinement requests, and reports are bound by SHA-256.

Table 3. Code and data inventory

Artifact	SHA-256
blocks	634d0d3302507a29cdbc5d358676ffc5c5fb3225696d5dbde053b657d8686f5e
stream_set	d0c989c7d6b9cd8f80679c71cb170b1b1616b58a2cfb8aca5ebd81697f06ab36

Hashes bind the report to frozen inputs.